Phase-Locking Without Prediction

A Pre-Registered Test of Kuramoto Crowding on Real Speculative Positions

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Abstract

Coupled-oscillator (Kuramoto) models are increasingly proposed as crowding indicators: when trading strategies phase-lock, synchronized exits should precede drawdowns. We document three implementation flaws that invalidate such evidence. Event-anchored phase extraction that interpolates between flow peaks uses the next peak and hence the future; a spread-based liquidity elasticity built from the high-low range regresses volatility on volatility; and i.i.d. p-values on overlapping forward drawdowns are the decisive one: on simulated markets with no predictability at all, one replication in five yields $p < 10^{-6}$. We then run a pre-registered test on real speculative positions: CFTC leveraged-fund and managed-money net positions in 32 futures markets, 2006–2026, with a causal phase. Speculative positioning does phase-lock across markets far beyond a circular-shift null (mean order parameter 0.287 against 0.156, $p = 0.0005$), and the order parameter is nearly orthogonal to correlation-based crowding measures. Yet it does not predict 4-week returns, drawdowns or volatility of the speculative book after controls (Romano–Wolf adjusted $p \geq 0.33$), in or out of sample. Synchronization of real positioning is real; its use as a crowding warning is not supported.

Keywords: crowding, synchronization, Kuramoto model, Commitments of Traders, look-ahead bias, pre-registration. **JEL:** G11, G14, G17, C58.

1 Introduction

Crowding, the convergence of many investors on the same positions, is a recognised channel of systemic fragility (Pedersen, 2009; Brunnermeier and Pedersen, 2009). Holdings-based measures arrive with a lag; return-based measures capture the consequences rather than the crowding itself. Synchronization models from statistical physics offer an appealing alternative: map each strategy to a phase oscillator, measure the Kuramoto order parameter $r(t) = |N^{-1} \sum_i e^{i\theta_i(t)}|$, and read high r as crowding (Strogatz, 2000; Acebrón et al., 2005).

*Independent researcher, Geneva. arjunkathiravelupro@gmail.com. Code, data pointers, the frozen pre-registration and every result file are in the replication package. An AI coding assistant was used to write and test code and to draft text; the author checked every number against the result files and is responsible for the content.

This note does two things. First, it documents three flaws that invalidate predictive evidence for such a measure, and shows by simulation which of them actually manufactures significance (Section 2). We found all three in our own earlier, unpublished, rejected work; we state them generally because each is easy to reproduce. Second, it runs the test the crowding claim actually requires: synchronization of *real* speculative positions, measured causally, with a pre-registered predictive test (Section 3). The pre-registration, with its decision rule, was committed before any outcome was computed.

The result is a clean split. Real speculative positioning phase-locks across futures markets well beyond chance, at times one can name (late 2008, the QE2 reflation into May 2011, the end of 2015, early 2021). The order parameter carries information that correlation and principal-component crowding measures do not. But it does not forecast the losses, drawdowns or volatility that a crowding warning should forecast.

2 Three pitfalls

2.1 Event-anchored phases that see the next event

A popular way to extract a phase from a non-oscillatory flow series is to anchor it to events: a buying burst is assigned phase 0, the next selling burst phase π, and the phase is interpolated linearly in between. The interpolation needs the date of the *next* burst. At date t the phase therefore encodes the time remaining until the strategy reverses, which is exactly the information a forward-drawdown regression rewards. Figure 1 shows it on a single position series with irregular holding periods: the interpolated phase reaches π precisely at each future reversal, while a causal phase built from past bursts only, extrapolated at the median past half-cycle, cannot.

A direct check is simple and should be routine: alter the input after date τ and verify that no output before τ changes. On a noisy sinusoidal position the interpolating implementation fails it (11 phases before $\tau = 350$ change when the input after τ is flipped); the causal one passes. Peak detection that compares q_t with q_{t+1} adds a further one-step leak.

Does the leak bias the statistic reported in this literature? We simulate 20 markets with independent, zero-mean returns and log-AR(1) stochastic volatility over 3,000 days, so that nothing is predictable, run six trend strategies per market (four time-series momentum lookbacks and two moving-average crossovers), and compute the Spearman correlation between $r(t)$ and the 21-day forward maximum drawdown of the equal-weight strategy portfolio, in 200 independent replications (Table 1). The leak does not shift this particular statistic: the mean correlation is -0.0004 with the interpolating phase and -0.008 with the causal one, with the same dispersion. The look-ahead is a validity violation that any predictive use must remove, but in this design it is not what produces significant correlations. The inference is (Section 2.3).

Table 1: Spearman correlation between $r(t)$ and the 21-day forward maximum drawdown on 200 simulated universes with no predictability (20 markets, 120 trend strategies, 3,000 days). p is the standard i.i.d. rank-correlation p-value.

Phase	mean ρ	s.d. ρ	range of ρ	share $p < 0.05$	share $p < 10^{-6}$
Interpolating (event-anchored)	-0.0004	0.070	$[-0.18,\ 0.19]$	0.64	0.18
Causal (past events only)	-0.008	0.072	$[-0.17,\ 0.18]$	0.59	0.23

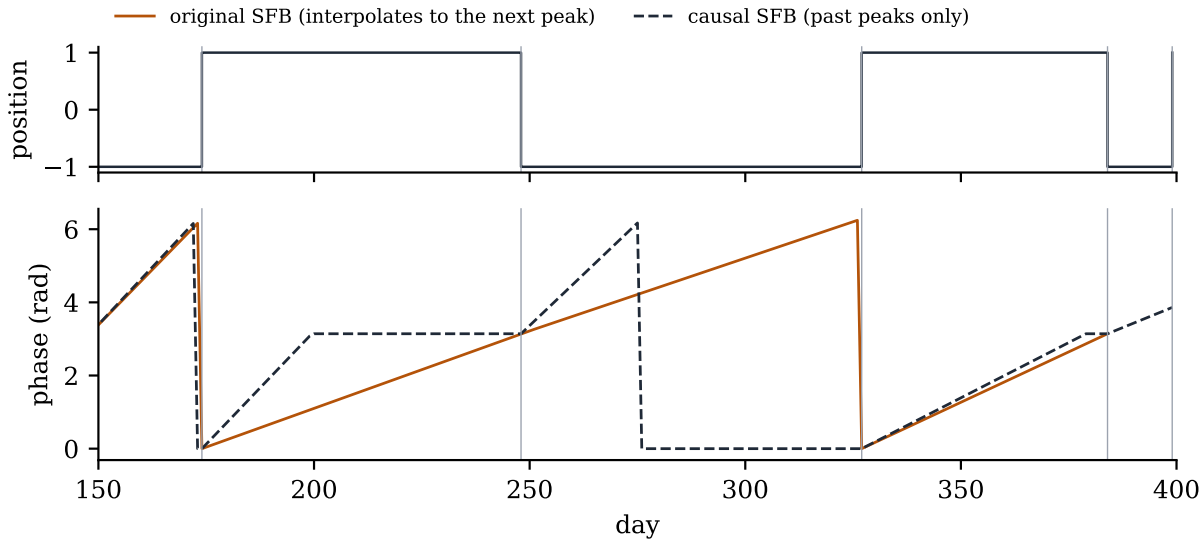


Figure 1: Event-anchored phase on a position with irregular holding periods. The interpolating phase (solid) reaches π and 2π exactly at the next reversal, which it cannot know at the time; the causal phase (dashed) uses past bursts only and must guess the duration.

2.2 A spread elasticity that is volatility on volatility

Crowding models often rest on a liquidity elasticity β, estimated from $\log V_t = \alpha - \beta \log \sigma_t + \varepsilon_t$ with V_t the inverse of a spread. When the spread is the Corwin and Schultz (2012) estimator, V_t is a function of the daily high-low range, itself a volatility estimator, and β is mechanically positive. We simulate a market whose true spread is constant, so that liquidity never responds to volatility and the true β is zero, and apply the estimator with a 21-day close-to-close σ_t. The estimate is $\hat{\beta} = 0.87$ ($R^2 = 0.25$) with a zero spread, 0.81 with 2 bp and 0.71 with 10 bp: values typical of those reported on real futures (a mean of 0.73 across 43 CME contracts in our earlier work). Quote-based elasticities are not immune either: in our tick data, replacing contemporaneous realized volatility by its one-day lag, to remove bid-ask bounce, leaves a rank correlation of only 0.24 with the baseline estimates (24 of 43 signs change); a 5-day median leaves 0.05.

2.3 i.i.d. inference on overlapping horizons

A 21-day forward drawdown sampled daily overlaps with its 20 neighbours, and $r(t)$ is itself persistent (weekly AR(1) of 0.77 on real positions). A standard rank-correlation p-value treats these observations as independent. Table 1 shows what that does when nothing is predictable: 59–64% of the replications are "significant" at 5%, and 18–23% reach $p < 10^{-6}$. The null correlations have a standard deviation of 0.07 and reach ± 0.19: in this design a correlation near 0.2 is roughly a three-standard-deviation event, not a one-in-a-million one, and our earlier headline figure ($\rho = 0.196$, reported with $p < 10^{-6}$, from a larger but similarly overlapping design) sits at the edge of what pure noise produces here. Every inference below uses a stationary block bootstrap (Politis and Romano, 1994) and Newey–West standard errors.

3 A pre-registered test on real positions

3.1 Data and design

Positions are the weekly CFTC Commitments of Traders, futures only, Tuesday as-of: leveraged funds in the Traders in Financial Futures report for equity indices, Treasuries and currencies, managed money in the disaggregated report for energy, metals, grains and livestock. The net position of market i is $x_i(t) = (\text{long} - \text{short})/\text{open interest}$. A report is used from the first trading day on or after the Monday following its Tuesday as-of date (it is released on Friday afternoon), and report weeks around the 2013, 2018–19 and 2025 government shutdowns, when releases were late, are excluded. The universe is every market with at least 85% coverage of both the CFTC series (2009–2026) and our daily continuous futures (2010–2026): 32 markets.

The primary phase is causal and needs no event detection. With $z_i(t)$ the 156-week trailing z-score of x_i and $d_i(t)$ the trailing z-score of its weekly change,

$$\theta_i(t) = \operatorname{atan2}\big(-d_i(t),\, z_i(t)\big), \tag{1}$$

so that 0 is a maximal long, $\pi/2$ selling out of it, π a maximal short and $3\pi/2$ buying. Two secondary variants are reported without decision weight: the nematic order parameter $|N^{-1}\sum_i e^{2i\theta_i}|$, which treats long and short extremes as aligned, and the causal version of the event-anchored phase of Section 2.

H1 (synchronization exists): the mean of $r(t)$ exceeds the distribution obtained by circularly shifting each market's phase series by an independent random offset (2,000 surrogates), which keeps each market's own dynamics and destroys cross-market alignment; threshold $p < 0.01$.

H2 (synchronization precedes unwinds): in weekly regressions with 4-week horizons (770 weeks, entries 2010-08 to 2026-01), $r(t)$ predicts at least one of: Y1, the return of the speculative book (each market long or short with the sign of x_i, volatility-scaled, equal weight); Y2, its maximum drawdown; Y3, the cross-market log ratio of forward to trailing realized volatility. Controls: the book's trailing 4-week return, the trailing volatility ratio, the mean pairwise correlation of position

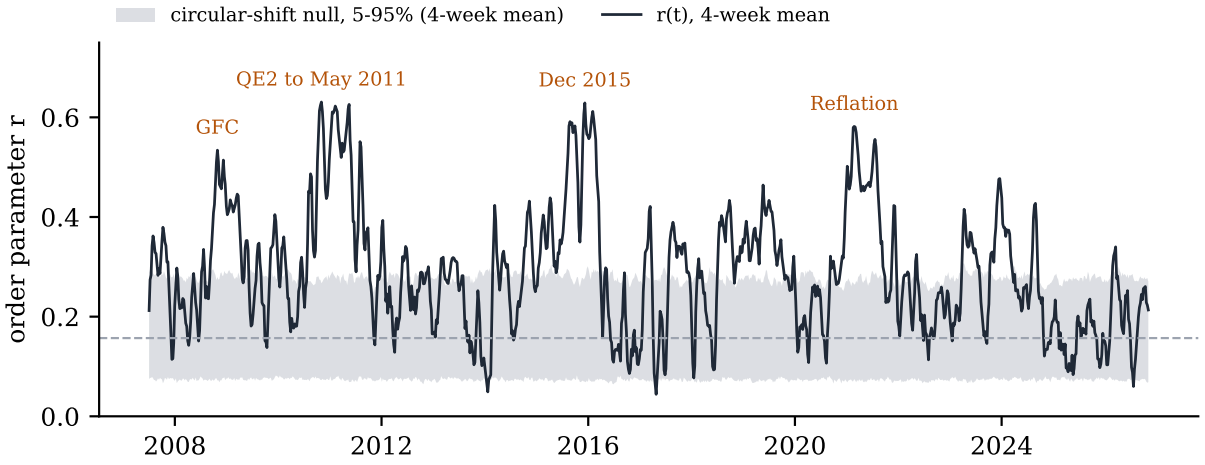


Figure 2: Order parameter of speculative positioning across 32 futures markets (causal phase, 4-week mean) against the 5–95% band of 500 circular-shift surrogates.

changes over 13 weeks, the first-principal-component share of position changes over 26 weeks, and the mean absolute position z-score. Inference: stationary block bootstrap (mean block 26 weeks, 5,000 draws) with a Romano–Wolf step-down across the three outcomes (Romano and Wolf, 2005); predicted signs $b < 0$ for Y1 and Y2, $b > 0$ for Y3.

H3 (out of sample): fit on entries to 2017, forecast 2018–2026, and test against the model without $r(t)$ (Clark and West, 2007).

The decision rule, fixed in advance: if H1 fails, the crowding claim is abandoned; if H1 holds and H2 fails, synchronization is descriptive only; a full crowding paper requires H1, H2 and H3.

3.2 Results

H1 holds. The mean order parameter is 0.287 against a surrogate mean of 0.156 (99th percentile 0.181; ratio 1.84; $p = 0.0005$, the smallest attainable with 2,000 surrogates). The surrogate mean equals the i.i.d. floor $\sqrt{\pi/4N} = 0.157$, so the shifted series behave as independent phases. Synchronization holds within financial markets (0.312 against 0.228) and within commodities (0.368 against 0.225), and with the causal event-anchored phase (0.211 against 0.156). Figure 2 shows that it is episodic: r leaves the null band in late 2008, from the QE2 reflation trade into the commodity sell-off of May 2011, at the end of 2015 and in early 2021. The order parameter is nearly orthogonal to standard crowding proxies: its Spearman correlation with the pairwise correlation of position changes is -0.05, with the first-component share -0.05, with the mean absolute position z-score 0.10. It measures alignment in the position-and-flow plane, which correlation of flows alone does not see.

Table 2: Pre-registered predictive tests (H2, H3). Standardized coefficient b on $r(t)$ with controls; t from Newey–West (8 lags); p_{RW} Romano–Wolf adjusted from 5,000 stationary-bootstrap draws; Clark–West one-sided p and out-of-sample R^2 relative to the model without $r(t)$ (training to 2017, 396 test weeks).

Phase	Outcome	b	t_{NW}	p_{RW}	p_{CW}	R^2_{oos}
Primary	Y1 return	0.054	0.71	0.78	0.40	0.000
	Y2 drawdown	−0.080	−1.58	0.33	0.21	0.005
	Y3 vol. spike	−0.002	−0.27	0.78	0.82	−0.006
Nematic	Y1 return	−0.128	−1.91	0.06	0.08	0.007
	Y2 drawdown	−0.100	−2.22	0.02	0.13	0.006
	Y3 vol. spike	0.000	−0.01	0.99	0.56	0.000
Causal events	Y1 return	−0.078	−1.28	0.39	0.07	0.002
	Y2 drawdown	0.002	0.05	0.96	0.51	0.000
	Y3 vol. spike	0.003	0.35	0.92	0.80	−0.008

H2 and H3 fail. Table 2 reports the predictive tests. With the primary phase, no outcome comes close: adjusted p-values of 0.78, 0.33 and 0.78, and out-of-sample R^2 below one percent. The nematic variant, which counts simultaneous long and short extremes as aligned, has the predicted sign for drawdowns ($p_{RW} = 0.02$) but does not survive out of sample ($p_{CW} = 0.13$) and carries no weight under the pre-registered rule; we report it as a lead for a separate, newly registered test.

A disclosed first run. Our first run loaded only the 2006–2016 financial archive because of a file-name pattern, and its universe was the 16 commodity markets. On that sub-universe the primary order parameter predicts 4-week returns of the speculative book with the *opposite* sign to the crowding thesis ($b = 0.33$, $t_{NW} = 3.20$, $p_{RW} = 0.004$; out of sample $p_{CW} = 0.02$): synchronized commodity positioning was followed by continuation, not unwinding. The bug was fixed, documented in a dated amendment, and the 32-market run is the registered result; we mention the commodity result because it shows how much a single universe choice can change the conclusion.

4 Conclusion

Real speculative positions phase-lock across futures markets, in identifiable episodes, and the Kuramoto order parameter captures an alignment that correlation-based crowding measures miss. None of this translates into a warning: in a pre-registered design with point-in-time data, synchronization predicts neither losses, nor drawdowns, nor volatility of the speculative book. Studies that report the opposite should show that their phases are causal (an alteration test costs one line), that their liquidity elasticities are not built from volatility, and that their inference respects overlapping horizons.

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